Project Demo Planning

Date	02 JULY 2026
Team ID	XXXXXX
Project Name	**OptiCrop: Smart Agricultural Production Optimization Engine**
Maximum Marks	1 Mark

Project Demo Planning:

A well-structured demo plan ensures that the team presents the project effectively, covering all key aspects in a clear and organized manner. This document outlines the plan for demonstrating the project, including the flow of the demo, key features to highlight, and responsibilities of each team member.

S.No	Demo Section	Description	Duration (mins)	Responsible Member
1	**Problem Intro & Dataset Setup**	Briefly present the agricultural problem and Kaggle dataset (Crop Recommendation Dataset) including NPK, temperature, humidity, pH, rainfall features.	2 min	jyothishmathi
2	**Data Engineering & Preprocessing**	Show dataset cleaning, null handling, scaling, train test split, and seasonal crop grouping logic.	3 min	Jayabhargavi
3	**Exploratory Data Analysis (EDA)**	Present visual insights like nutrient distribution, rainfall patterns, and crop-wise correlations using graphs.	2 min	nyshita
4	**Model Training & Evaluation**	Explain ML models (Logistic Regression, KNN, Decision Tree, Random Forest, K-Means) and performance comparison.	3 min	Jayabhargavi
5	**Live Web Application Demo**	Show Flask web app, input soil values, and demonstrate real-time crop prediction output.	3 min	nyshita
6	**Conclusion & Insights**	Summarize system benefits, accuracy insights, and real-world agricultural impact.	2 min	jyothishmathi

Demo Flow Summary:

Step	Activity	Notes
1	Introduction & Problem Statement	Explain need for smart crop recommendation to improve farming decisions and productivity.
2	Solution Overview	Present system architecture: ML pipeline, preprocessing, trained models, and Flask web app.
3	Live Feature Demonstration	Enter sample soil values and show real-time crop prediction output from Flask application.
4	Q&A Session	Answer questions on dataset features, model selection, accuracy, and real-world usage in agriculture.